

Plant Protein Classification using K-mer Encoding

A Report submitted

in partial fulfillment of the requirement

for the award of a Degree of

Bachelor of Technology

in

Computer Science and Engineering

by

M. Pravallika Devi
(2019BCSE039)

S. Pranitha Reddy
(2019BCSE032)

Under the guidance of

Dr. Veningston K



Department of Computer Science and Engineering

National Institute of Technology, Srinagar,

Kashmir 190006, INDIA

June 2023

CERTIFICATE

This is to certify that the project report entitled **Plant Protein Classification using K-mer Encoding** submitted by **Pravallika Devi M (2019BCSE039)** and **Pranitha Reddy S (2019BCSE032)** to the Department of Computer Science and Engineering, National Institute of Technology Srinagar, Kashmir, in partial fulfillment for the award of the degree of **B.Tech in Computer Science and Engineering** is a bona fide record of project work carried out by them under my supervision.

Dr. Veningston K

Supervisor

Department of Computer Science and Engineering

National Institute of Technology Srinagar

STUDENT DECLARATION

We declare that this project report titled **Plant Protein Classification using K-Mer Encoding** submitted in partial fulfillment of the degree of **B.Tech in Computer Science and Engineering** is a record of original work carried out by us under the supervision of **Dr. Veningston K.** The matter embodied in this project, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma. We also declare that the work submitted by us is entirely original, free from plagiarism, and has been diligently checked through Turnitin software to ensure its authenticity.

Pravallika Devi M

2019BCSE039

Pranitha Reddy S

2019BCSE032

Srinagar, 190006

13-06-2023

ACKNOWLEDGEMENTS

We consider it as our privilege to express gratitude and respect to all those who guided and inspired us in the completion of our B.Tech project. First of all, we would like to acknowledge and extend our heartfelt gratitude to our guide, Dr. Veningston, Assistant Professor at the Department of Computer Science and Engineering, National Institute of Technology, Srinagar for his valuable guidance, constant encouragement, and kind help at various stages for the execution of this dissertation work. An erudite teacher, a magnificent person, and a strict disciplinarian, we consider our self fortunate to have worked under his supervision.

We would also like to express our sincere thanks to the Department of Computer Science and Engineering at NIT Srinagar for providing us with this opportunity to work on this important project.

We also place on record our sense of gratitude to one and all, who directly or indirectly, have lent their hand in this venture.

Pravallika Devi M

Pranitha Reddy S

ABSTRACT

Proteins play an important role in human body and in plants. Lack of expertise in protein labeling in plants can make it extremely difficult to characterize and comprehend the precise roles and activities of different proteins. Furthermore, it restricts development in fields like biotechnology , disease resistance, and crop enhancement.

The presented project focuses on plant protein classification, aiming to overcome the challenges arising from limited protein labeling knowledge. Advanced machine learning techniques, including various classification algorithms such as Logistic Regression, Decision Tree, K-nearest neighbors (KNN), Support Vector Machines (SVM), Random Forest (RF), Multinomial Naive Bayes (NB), AdaBoost, and XGBoost, are employed to accurately classify protein sequences into their respective families. This classification approach provides valuable insights into the functions and roles of proteins within plants, ultimately advancing our understanding of plant biology. This attempt offers new possibilities for advancement in critical sectors such as agriculture, drug discovery, and genomic research by eliminating the limitations associated with limited protein labeling knowledge.

TABLE OF CONTENTS

DESCRIPTION	PAGE NUMBER
CERTIFICATE	iii
DECLARATION	v
ACKNOWLEDGEMENTS	vii
ABSTRACT	ix
LIST OF FIGURES	xiii
LIST OF TABLES	xv
ABBREVIATIONS/ NOTATIONS/ NOMENCLATURE	xvii
1. INTRODUCTION	1
1.1 Overview	1
1.1.1 Gene Expression	2
1.1.2 Genome Annotation	4
2. LITERATURE SURVEY	7
3. PROPOSED METHODOLOGY	19
3.1 Problem Statement	19
3.2 Dataset Description	19
3.3 Detailed Architecture	22
3.3.1 Dataset Preparation	22
3.4 K-mer Encoding	26
3.5 Algorithms Used	27
4. RESULTS AND DISCUSSION	33

4.1	Evaluation Metrics	33
4.1.1	Confusion Matrix	33
4.1.2	Accuracy	33
4.1.3	Precision	33
4.1.4	Recall	33
4.1.5	F1-score	34
4.2	Experimental Evaluation using Entire Dataset (using 34 classes)	34
4.3	Divided Datasets Evaluation	38
5.	CONCLUSION and FUTURE WORKS	47
	REFERENCES	49
	Appendix A Output	53

LIST OF FIGURES

FIGURE	TITLE	PAGE NUMBER
1.1	DNA Structure	2
1.2.	Central Dogma	3
1.3.	Genome Annotation	4
1.4.	General Architecture	5
3.1.	Plant GDB - Homepage	20
3.2.	Peach Genome	20
3.3.	Detailed Workflow	21
3.4.	peptide File Conversion (.fa to .csv)	22
3.5.	Annotation File Conversion (.txt to .csv)	23
3.6.	Final Dataset (filtered_data.csv)	23
3.7.	4-mer representation	26
4.1.	Evaluation metric scores of all models	34
4.2.	Confusion Matrix of all classes	37
4.3.	Accuracy plots [D1] [D2]	38
4.4.	Precision plots [D1] [D2]	39
4.5.	Recall plots [D1][D2]	39
4.6.	F1-score plots [D1][D2]	40
4.7.	Confusion Matrix of all classes [D1]	41
4.27.	Confusion Matrix of all classes [D2]	44

LIST OF TABLES

TABLE	TITLE	PAGE NUMBER
2.1	Summary of Literature Survey	14
3.1.	Class Label Description	24
3.2.	Sequence Encoding Methods	26
4.1.	Evaluation Metric Results	45
4.2.	Evaluation Metric Results - D1	45
4.3.	Evaluation Metric Results - D2	45

ABBREVIATIONS/NOTATIONS/NOMENCLATURE

Abbreviations

AdaBoost - Adaptive Boosting

BOW - Bag of Words

DNA - Deoxyribonucleic Acid

DT - Decision Tree

FN - False Negative

FP - False Positive

KNN - K Nearest Neighbor

LR - Logistic Regression

mRNA - Messenger Ribonucleic Acid

Multinomial NB - Multinomial Naive Bayes

PlantGDB - Plant Genome Database

RF - Random Forest

RNA - Ribonucleic Acid

SVM - Support Vector Machine

TN - True Negative

TP - True Positive

tRNA - Transfer Ribonucleic Acid

XGBoost - Extreme Gradient Boosting

Notations

A - Adenine

T - Thymine

G - Guanine

C - Cytosine

U - Uracil

Biological Nomenclature

Prunus persica - peach

CHAPTER 1

INTRODUCTION

1.1 Overview

Proteins are macromolecules that play crucial roles in the body. They are used to structure the body's organs and help in regulation and body functioning. They composed of hundreds of amino acids that are covalently connected to each other. There are amino acids of 20 types, and the sequence predicts the protein's 3D structure and corresponding function. Combining three nucleotide bases ('A', 'T', 'G', and 'C') results in coded amino acids.

Proteins play crucial role in plant growth and development. They provide a variety of functions, including photosynthesis, biosynthesis, transportation, immunology, etc. A diverse range of proteins, including enzymes, structural proteins, and storage proteins, are produced by plants.

- Enzymes – Act as catalysts and are crucial for plant metabolism.
- Structural Proteins – Provides support and shape to plant cells.
- Storage proteins – Stores amino acids for later use.

DNA is a macromolecule composed of two polynucleotide chains that forms double helix structure by coiling around each other as shown in fig 1.1. It carries the genetic code, which is essential for all living things, to develop, function, grow, and reproduce. The two DNA strands are made up of nucleotides further referred to as polynucleotides. Each nucleotide is made up of a phosphate group, a deoxyribose sugar, and one of the four nitrogen-containing nucleobases (A, T, G and C).

Since more than 98% of human DNA is non-coding, it cannot serve as a guide for the construction of protein sequences. Due to the fact that they move in the opposing directions, the two DNA strands are antiparallel. Transcription is the process of converting DNA nucleotides into RNA strands, with the exception of thymine (T), for which RNA substitutes uracil (U). Translation is the process by which these RNA strands use the genetic instructions to direct the arrangement of amino acids in proteins.



Fig 1.1 DNA Structure

1.1.1 Gene Expression

The genetic information of a cell is stored in DNA or RNA chemical form.

- A Gene is a Portion of a DNA molecule.
- The Genome is the entire DNA in a cell.

Each cell has a complex and closely controlled process from gene to protein. Transcription and translation are the two main phases. Transcription and translation work together to produce gene expression.

The information contained in a gene's DNA is transferred to a similar molecule called RNA (ribonucleic acid) in the cell nucleus during the transcription process. Since messenger RNA (mRNA) transmits information from the nucleus to the cytoplasm, it is this type of RNA that contains the instructions needed to make a protein.

The cell wall is where translation, the second step in turning a gene into a protein, takes place. The ribosome and the mRNA interact, and the ribosome "reads" the nucleotide sequence of the mRNA. Three nucleotide sequences make up a codon, which typically codes for one amino acid. One amino acid at a time, transfer RNA (tRNA), a kind of RNA, assembles the protein. Until the ribosome encounters a "stop" codon (a three-nucleotide sequence that does not code for an amino acid), protein production continues.

The transmission of information from DNA through RNA to proteins is one of the fundamental ideas in molecular biology ("central dogma").

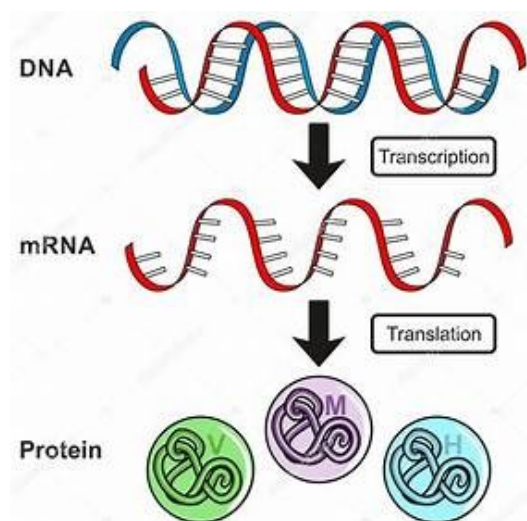


Fig 1.2 Central Dogma

1.1.2 Genome Annotation

Giving the sequence of the genome is never enough we need to know which part of the genome is coding and which part is non-coding. Here comes Genome Annotation, which is the process of describing the structure and function of a genome. Gene sequencing is the process of determining the nucleic acid sequence, or the arrangement of the nucleotides in DNA.

Annotating a genome consists of three major steps:

- Identifying the genome regions that do not contain protein-coding genes.
- Gene Prediction - Process of identifying genome elements.
- Linking biological information to the elements

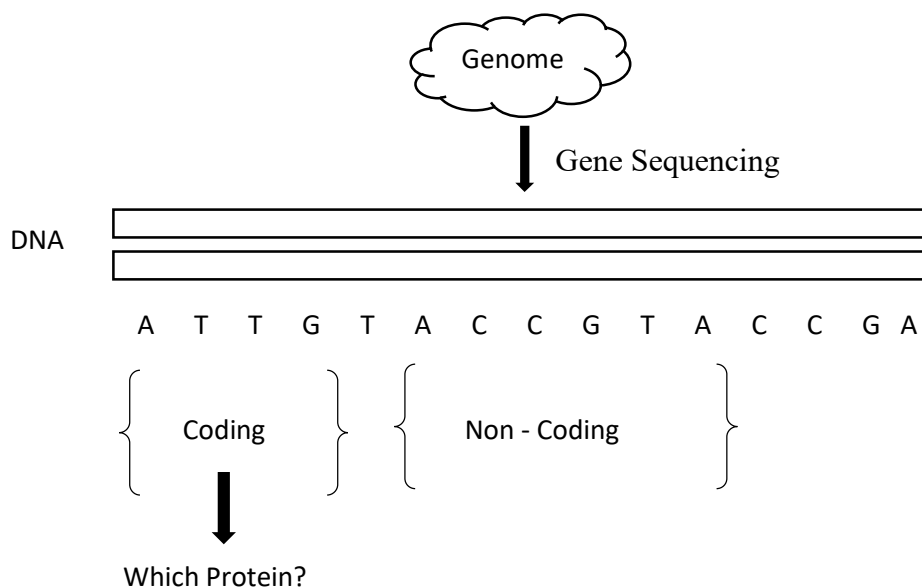


Fig 1.3 Genome Annotation

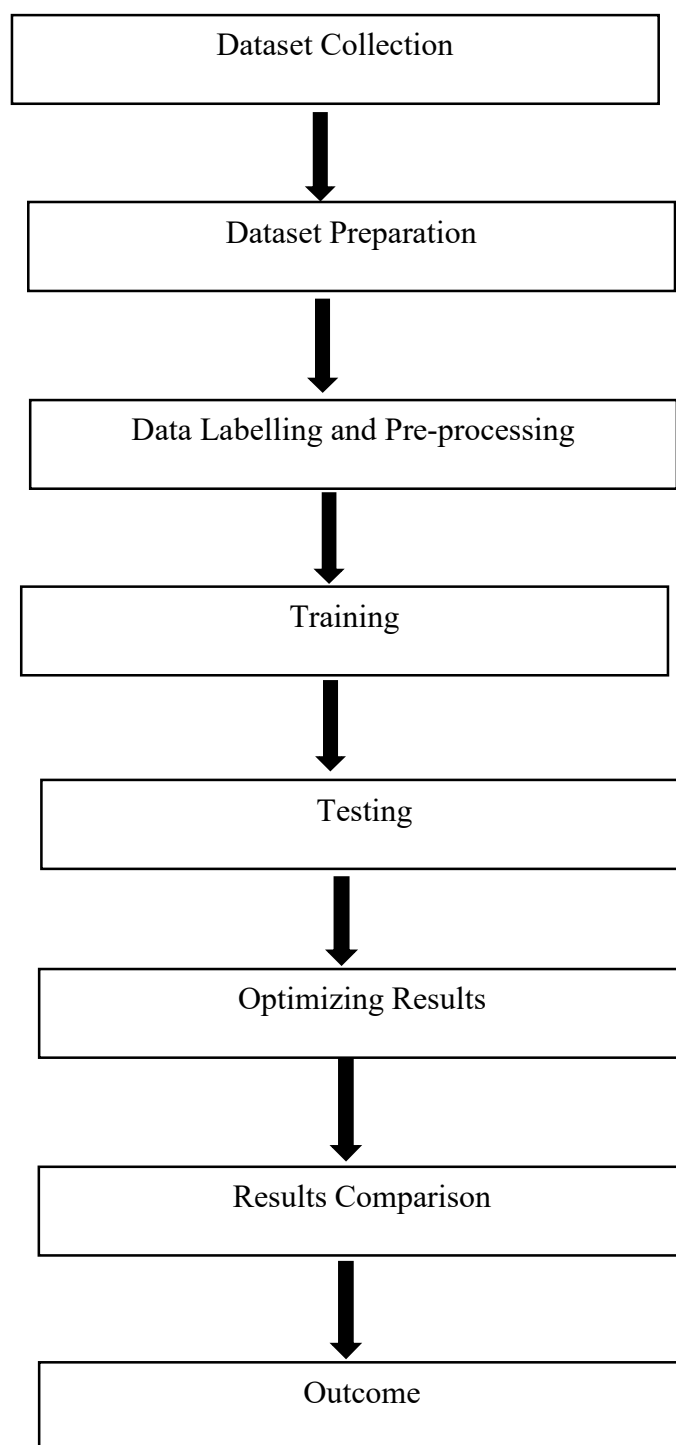


Fig 1.4 General Architecture

CHAPTER 2

LITERATURE SURVEY

1. Protein classification using genome neighborhood.

This researcher created a computational strategy employing machine learning (ML) and genomic neighborhood information to find photosynthetic protein and taken UniprotKB/Swiss-Prot database [12]. Genome neighborhood network (GNN) outperformed Random forest, decision tree methods with an accuracy of 87%. As evidence of the model's potential to increase photosynthetic efficiency, it also showed its capacity to recognize novel photosynthetic proteins. The key outcome of this is the functional relationship between the neighbors of photosynthetic genes.

2. k-mer - HIV - 1 Genome Subtyping

The study focuses on the classification of virus subtypes, notably within HIV-1. To address this issue, the researchers suggest using an open-source technique called Kameris, which classifies HIV-1 subtypes based on k-mer frequencies [17]. Kameris outperformed other evaluated software, achieving a high accuracy of 97% while processing more than 1,500 sequences per second. 15 classifiers were trained during supervised classification utilizing k-mer frequencies as feature vectors. SVM, LR, decision trees, and RF were utilized. The effectiveness of the categorization process was evaluated using cross-validation. The technique calculated pairwise distances between sequences based on their k-mer frequencies.

3. Evaluating Plant Gene Models Using ML

Truegene is an ml approach developed for gene models classification and reduce false positive annotations in gene prediction [4]. The Legume Information System (LIS) provided the Pisum annotated gene and protein dataset. The NCBI NR database was

compared to the annotated genes. It makes use of 41 protein-based traits, including amino acid and nucleotide sequences, and 14 genes. The Pisum genome was used for XGBoost model training, which resulted in optimized models with 87-90% prediction accuracy and F-1 scores of 0.91-0.94.

4. LTR retrotransposons Classification using K-mer method

Although LTR retrotransposons are prevalent repeating sequences in plant genomes, the classification often involve laborious, manual procedures [7]. To overcome the problem, the researchers created a technique for classifying LTR retrotransposons and mapping them into certain families using K-mer based ML algorithms. They used InpactorDB, which contains 67,241 LTR retrotransposon sequences from 195 plant species that have been systematically categorized into families. This dataset included sequences from Repbase, RepetDB, and PGSB. Their approach achieved an impressive of 95% F1-Score.

5. Classification of plant transcription factor proteins

Science still faces difficulties in categorizing amino acid sequences and comprehending the connection between amino acids and protein synthesis. To categorize the amino acid sequences of unidentified species, this study develops a hybrid DLM that makes use of the plant transcription factor database [1]. The model successfully classified transcription factor proteins in the kingdom of plants with a high success rate of 98.23% throughout tests. The hybrid model performs better than traditional LSTM- or CNN-based models due to its lightweight layers and shorter training time. The success of the suggested model is further increased by the usage of Word2Vec vectors.

6. Plant vacuole protein prediction based on sequence

Prediction of subcellular localization is essential for comprehending gene functions in proteomes. This study addresses this issue by developing various compositions and

PSSM-based models, resulting in improved accuracy compared to previous methods [13]. They used UniportKB/SwissProt database. The best model achieved approximately 63% accuracy on a blind dataset, surpassing current tools. To make the models accessible, they developed 'VacPred' [GUI based software], compatible with Windows and Linux platforms. They got an accuracy of 86.49%/87.84% and a sensitivity of 90.54%/93.24%.

7. Protein Sequence Classification in Human using ML methods

Various methods, including sequence similarity and ML approaches, have been used for protein function prediction. Swiss-Prot is the dataset (reviewed), whereas TrEMBL is unreviewed [15]. There are 559228 proteins in the reviewed data set and 146106279 in the unreviewed dataset. This article compares six classification techniques (CART, Quick unbiased Efficient Statistical Tree, CHAID, C5.0, ANN, and SVM) and finds that C5.0 and SVM methods outperformed with an accuracy above 94%. The researchers analyzed a dataset of 4,314 proteins, each with 1,438 features.

8. DNA Sequence Classification in ML: K Mer Counting

In genomic research, DNA sequences need to be classified to determine the applicability of a new protein. This study utilizes machine learning algorithms to identify classes of DNA sequences based on nucleotide sequences [5]. Open DNA sequence dataset was used to obtain the gene sequence dataset. Different datasets representing gene families are analyzed using substrings of defined lengths (determined by the 'k value') to capture sequence patterns. When they took the human dataset $k=6$ they obtained the best accuracy 98.4%, and Precision 98.4%. The researchers obtained a gene sequence dataset from Kaggle containing three text files: chimpanzee, dog, and human.

9. Plant Protein Classification using Ensemble Classifiers

The method for forecast the subcellular localization using several classifiers is enhanced in this research. The authors suggested an ensemble ML approach based on average voting [8]. The dataset for testing and training was obtained from Plant-mSubP. They gather different features appropriate for each sort of localization, use feature selection to lessen dimensionality, and then train three different models. According to experimental findings, using the testing dataset, the suggested ensemble technique could correctly classify objects in 11 compartments with an accuracy of 84.58%.

10. DNA Sequencing Using ML Algorithms

DNA sequencing is an essential technique for discovering a person's genetic makeup and comprehending variations and mutations linked to genetic illnesses [6]. This data aids in understanding the patient's health status and aids in the planning of potential treatments for the patient. In this research, DNA is sequenced using a human dataset utilizing ML approaches like SVM, CNN, LSTM, RF, AdaBoost, Naive Bayes, and K-Nearest Neighbour (KNN). Finally, the LSTM model performed the best with a test accuracy of 99.5%.

11. Plant electrical signal classification using NN model.

Graph neural network model called PMESP was proposed in this study to discover electrical signal-related proteins. GNN model constructs a graph structure for visualizing protein interaction relationships and utilizes a Bi-LSTM for electrical signal feature extraction [2]. They used CNN model to predict proteins and a convolutional encoder for link prediction. They designed a dataset named PesPPI which comprises of 23,214 PPIs pairs and 3600 nodes. As shown by the results, the values of the model's AUC curve and the ROC curve are 96.24% and 96.84%, which gave better performance results than earlier techniques.

12. Accuracy Prediction of a sample DNA Sequence Dataset

The article analyses the development of DNA sequencing approaches and contrasts current methods with earlier ones. The importance of DNA as the blueprint for life and its critical function in the growth and survival of all creatures are emphasized [6]. They used a set of exploratory datasets. It comprises of seven gene families and three organisms: humans, chimpanzees, and dogs. DNA sequencing has advanced significantly as a result of the quick rise in data generation. The New-Age DNA sequencing methods are significantly more effective than the older methods. They got 98.4% accuracy. Additionally, the promise of machine learning is highlighted, particularly when utilizing a Multinomial Naive Bayes classifier to predict DNA sequences.

13. DNA Sequence Classification using CNN

The classification of DNA sequences within the framework of the analysis of biomedical data was the study's main focus. The researchers used deep learning (DL) models, which have the ability to automatically extract pertinent features from the input data, to address this difficulty [9]. They specifically used three different architectures: CNN-LSTM, CNN-Bi LSTM, and CNN-Convolutional Neural Network (CNN). Both Label encoding and K-mer encoding strategies are used for representing the DNA sequences. When the models were tested using a variety of classification criteria, the CNN and CNN-Bi LSTM with K-mer encoding both demonstrated good accuracy, scoring 93.16% and 93.13% on the testing data.

14. K-mer - Fixed-length DNA sequencing

By using k-mer frequency and subsequences commencing at certain places as categorical features, the algorithm for categorizing fixed-length DNA sequences performed well [5]. They chose the datasets from the UCI ML Repository, Promoter, and Splice. Four models namely iNuc-PseKNC, iNuc-PseSTNC, 3LS and TNS were

used. K-mer method used in generating fixed-length vectors. In the human dataset, the prediction Accuracy is 86.33%.

15. Prediction of Hormone-Binding Proteins using NB

In order to correctly identify hormone-binding proteins (HBPs), a prediction model dubbed HBP_NB is put out in this paper. Traditional biochemical tests are expensive and time-consuming, making detecting HBPs difficult [10]. A collection of HBPs from the UniProt database is used in the HBP_NB model, and to extract features, k-mer approach is used. To reduce dimensionality and choose the best feature collection, feature selection is used. The chosen characteristics are then used to build the prediction model using Naive Bayes, which is assessed using Cross validation approach (cv=10). The accuracy, sensitivity, and specificity of the results—95.45%, 94.17%, and 96.73%—indicate the viability and efficacy of the HBP_NB model.

16. Protein Class Prediction on Count Vectorizer

Machine learning technologies are crucial for sifting through this data for patterns and characteristics that may be used for stress phenotyping [14]. This study offers classification of ML tools and best-practice guidelines for the plant community to successfully employ ML approaches in analyzing biotic and abiotic stress features. The performance of multinomial NB classifier, KNN, RF, DT, and LR with k-mer encoding on test data is 93.16% and 93.13%, respectively.

17. DNA sequencing Classification in bioinformatics

An important area of bioinformatics study is the use of AI algorithms to group DNA. Researchers employ a range of techniques, including multinomial NB, SVM, KNN, decision trees, random forests, and logistic regression, to categorize DNA sequences and discover novel genes [17]. The usage of label and k-mer encoding allows these models to reach an accuracy of about 93% on testing data. These approaches improve the management of biological information while assisting in identifying the components of proteins.

18. Plant Allergenic Protein Prediction based on sequence

This paper comprehends the allergenic properties of dietary proteins. By combining supervised and unsupervised machine learning approaches, it is possible to predict if plant proteins may cause allergic reactions [3]. The method comprises rating descriptors and evaluating the effectiveness of their categorization. An SVM (support vector machine) is utilized for partitioning, while a KNN (k-nearest neighbor) classifier is used for classification. For variable selection and final classification, cv (cv=5) method is used to validate the KNN classifier. In order to address food allergies, the study emphasizes the necessity for a reliable and practical protein classification system.

19. DL model for predicting next-generation sequencing

Researchers have created a DL model to forecast the depth of next-generation sequencing (NGS) [11]. In three separate Next-Generation Sequencing (NGS) panels, including human SNP, human lncRNA, and non-human sequences for storage of DNA information, Depth Sequencing was predicted using a DLM with 93% accuracy for Single nucleotide polymorphisms and 99% accuracy for non-human panels. Independent testing revealed that the DLM trained on SNP panel correctly predicted the lncRNA panel 89% of the time.

Table 2.1: Summary of Literature Survey

S.No	Technique & Reference	About Technique	Applications	Dataset Used	Results	Limitations
1	RF, DT, and GNN [12].	RF is an ensemble method; DT is classification algorithm, GNN is a gene analysis method.	Protein Classification using GNN.	UniprotKB/Swiss-Prot.	GNN Accuracy - 87%.	Since just photosynthetic bacterial proteins were used, the system functions well with prokaryotic proteins alone.
2	RF SVM, LR, DT [16].	All ML methods are used for classification or regression.	k-mer-HIV1 genome subtyping.	Los Alamos (LANL) sequence database.	Kameris accuracy - 97%.	This method does not offer a justification for how a DNA sequence is categorized.
3	XGBoost, SVM, DT [4].	Xgboost is an ensemble method. DT is tree-based method, svm is an ml method.	Evaluating Plant Gene Models Using ML.	Pisum annotated gene and protein dataset from LIS.	XGBoost classifier 87-90% accuracy.	—
4	LSV Classifier, LR, KNN, NB, DT, RF [7].	LR & SVM are linear classification methods and KNN is a classification algorithm.	k-mer method to classify LTR-retrotransposons in plant genomes.	inpactorDB contains 67,241 LTR retrotransposon sequences from 195 plant species.	F1-Score of 95% .	Reducing the number of features won't improve the performance of the algorithm because processing them can be expensive in terms of CPU power.

S.No	Technique & Reference	About Technique	Applications	Dataset Used	Results	Limitations
5	CNN, ResNet-based ProtCNN, LSTM [1].	CNN is a deep learning algorithm.	Plant transcription factor proteins Classification of DLM.	Plant transcription factor database.	It classified transcription factor proteins of plants with rate of 98.23%.	It get the three-stage hybrid model ready to use genetic resources, to perform more efficiently on larger datasets.
6	SVM, Five-fold cross-validation [13].	SVM method can be used for the classification or regression tasks.	Sequence-based prediction of plant vacuole.	UniportKB/SwissProt database.	K-PSSM and TRI-PSSM accuracy 86.49%/87.84%.	—
7	ANN, SVM, Classification and Regression tree [15].	All of these methods are supervised learning techniques and used for solving classification.	Human Protein Sequence Classification using ML.	Two datasets: Swiss-Prot (reviewed protein dataset), TrEMBL (unreviewed dataset).	Support Vector Machine accuracy of 94%.	Due to the imbalance in the dataset, these classes have poorer precision and recall values than other classes.
8	MNB, CNN, LR [5].	LR is linear model, CNN is used for image classification purpose.	DNA Sequencing Classification through ml.	Gene sequence dataset from Kaggle containing 3 text files: chimpanzee, dog, human.	Human dataset k=6 accuracy of 98.4%.	It cannot analyse natural data in its raw form as quickly as Deep learning can.
9	XGBoost, RF, KNN [8].	These techniques are come under ensemble learning.	Protein Classification in Ensemble Classifiers.	Plant-mSubP.	The ensemble method achieved an overall accuracy of 84.85%.	—

S.No	Technique & Reference	About Technique	Applications	Dataset Used	Results	Limitations
10	SVM, CNN, RFt, AdaBoost, NBs, KNN, LSTM [6].	RF is an ensemble model, NB is a probabilistic method, KNN is used for classification purposes.	DNA Sequencing using ml algorithms.		LSTM model performed the best with accuracy of 99.5%.	In addition to its conventional uses in medicine, and diagnosis, it can also be employed in forensics and agriculture.
11	CNN, KNN, NB [2].	CNN is deep learning algorithm, KNN is a classification algorithm, NB is a probabilistic method.	Plant electrical signal classification in biological mechanisms.	They designed a dataset named PesPPI, it has 23,214 PPIs Pairs and 3600 nodes.	recall curve values are 96.24% and 96.84%.	The PMESP can learn about characteristics of electrical sign data with different salt, which can truly reveal schemes of data.
12	MNB, RF [6].	RF is an ensemble method, NB is a Probabilistic method.	Accuracy Prediction of a sample DNA sequence.	exploratory datasets of seven gene families and three organisms.	They got 98.4% accuracy in multinomial naïve Bayes.	—
13	CNN, CNN-Bidirectional LSTM, CNN-LSTM [9].	The mentioned techniques are ml algorithms.	DNA Sequence Classification using CNN.	Public nucleotide sequence database.	CNN - 93.16%.	LSTM with label encoding is a solid choice for improving class recall and sensitivity.

S.No	Technique & Reference	About Technique	Applications	Dataset Used	Results	Limitations
14	RF, SVM [5].	RF is a ensemble model, SVM is method that can be used for classification	DNA Sequence Classification using K-mer.	They chose the datasets from UCI ML Repository of humans.	In human dataset accuracy is 86.33%.	In addition to other areas of sequence recognition, it can be used to classify protein sequence data.
15	NB, RF, LR [10].	NB use Bayesian probability for modeling task, LR is Linear model.	Prediction of Hormone-Binding Proteins.	From UniProt database, they collected hormone-binding proteins.	Hormone-Binding Proteins_ Naive Bayes accuracy of 94.17%.	Collection of training and test set trials required(due to tiny dataset).
16	SVM, KNN, RF, DT, LR, MNB [14].	SVM & LR are linear models, DT is a tree-based method, RF is an ensemble method.	Protein Class Prediction on Count Vectorizer.	They generated real-time data of remote-sensing , velocity.	k-mer encoding on test data they got 93.16%.	—
17	KNN, SVM, MNB, RF, LR, DT [17].	KNN & DT are used for classification purpose, NB is a probabilistic method.	DNA Sequencing classification in bioinformatics.	Public nucleotide sequence database.	SVM, KNN -- 93%.	—

S.No	Technique & Reference	About Technique	Applications	Dataset Used	Results	Limitations
18	SVM, KNN, NB, RF [3].	SVM & KNN used for classification, NB is a probabilistic method, RF is a ml method.	Prediction of Plant Allergenic Proteins: ML Classification.	They gathered 954 food allergens and nonallergens.	SVM accuracy of 93% compared to KNN, RF, and NB.	—
19	CNN, ANN, SVM, LR [11].	LR & SVM are linear models, ANN is a neural network, CNN is a DL algorithm.	Predicting next generation sequencing using DLM.	DNA interaction kinetics rate constants and NGS sequencing depth.	They got 93% for SNP and 99% accuracy for non-human panels.	Limited the inputs to a minimal set of local variables that can be automatically computed using DNA sequence.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Problem Statement

Build a robust classification model that correctly maps amino acid sequences to protein families within the PlantGDB Database [1]. In this study, several models were used to solve the above problem. The model analyzes the input sequence data and predicts the protein class it belongs to with high accuracy and precision utilizing Machine Learning algorithms. By proposing an efficient and automated approach for protein family classification, this work aims to improve knowledge of protein structure and function.

3.2 Dataset Description

PlantGDB [1] is a data repository containing different plant species' genome sequences. In addition to sequence data, it also provides alignments and annotations. 'Prunus persica [peach] genome' [2] taken from [1]. From [2], peptide file (Ppersica_139_peptide.fa.gz) and the annotation file (Ppersica_139_annotation_info.text.gz) were considered as shown in Fig 3.2.

PlantGDB ...resources for comparative plant genomics

Home Sequence Genomes Tools Datasets Outreach Help

PlantGDB
Home
Sequence
Download
Search
EST Assemblies
FTP Server
More...
Genomes
Genome Browsers
Annotation
Tools
BioExtract
BLAST
GeneScaper
More...
Datasets
AcDe Tagging
ASIP
SRGD
More...
Outreach
PGRIP
Native American
Help
Help & Tutorials

Welcome to plantgdb.org! Tools and resources for plant genomics
Quick tips for using PlantGDB...

Public Plant Sequence Release 187 is current.
Species-paired [Unifreeze](#) sequences from GenBank and UniProt.
More about Release 187...

Version Release 187 (2011-12-15)
Resources: [Search](#) [Download](#) [BLAST](#)
See below for totals by type. [View example](#) of a sequence record. [View all sequence resources](#) at PlantGDB.

Seq Type	Description	Species Count	Sequence Count
Protein	UniProt Protein	50,105	708,327
EST	Expressed Sequence Tag	870	23,579,550
OSS	Genome Survey Sequence	358	13,868,488
STS	Sequence Tagged Sites	273	150,889
HTG	High Throughput Genomic DNA	87	25,458
HTC	High Throughput cDNA	120	68,543
PLN	PLN Division mRNA	107,075	1,749,128

Genome Browsers
[Genome browsers](#) for emerging & completed plant genomes.
Video Tutorial (5 min): [View Quicktime](#) [View Flash](#)
More about Genome Browsers...

Genome Databases at PlantGDB

GDB	Species	Type	Assembly	GenBank
AtGDB	<i>Arabidopsis thaliana</i>	Chromosome	TAIR9	TAIR10
BdGDB	<i>Brachypodium distachyon</i>	Chromosome	Bd1	Bradi1.2
BrGDB	<i>Brassica rapa</i>	Chromosome		
CpGDB	<i>Citrus papaya</i>	Scaffold	GenBank	GenBank
CrGDB	<i>Chlamydomonas reinhardtii</i>	Chromosome	JGI4.0	JGI4.3
CuGDB	<i>Cucumis sativus</i>	Scaffold	JGI	JGI
GnGDB	<i>Gossypium hirsutum</i>	BAC	n/a	GenBank
GmGDB				

Figure 3.1: PlantGDB – Homepage (<http://plantgdb.org/>)

PeGDB Downloads

PeGDB encompasses primary sequence data, including the most current genomic sequence, EST and cDNA sequences from GenBank, EST and EST sequences onto the genome as well as interpretations of the overall data are based on our own spliced alignment results.

The links below, mirrored on our [FTP site](#), provide bzip2 archives of primary sequence used in the current version of PeGDB, as well as the sequences. For more information on the current datasets, their sources, and alignment methods, go to our [Data & Methods](#) page.

The peach genome assembly is still being refined and completed. PeGDB attempts to remain synchronized with the most recent release of the genome assembly. For more information or comments about this genome browser, please make use of our [feedback form](#).

Name	Last modified date	File Size
Ppersica_139.fa.gz	Nov 8 2010	63.77 MB
Ppersica_139_annotation_info.txt.gz	Nov 8 2010	610.19 KB
Ppersica_139_cds.fa.gz	Nov 8 2010	10.28 MB
Ppersica_139_gene.gff3.gz	Nov 8 2010	2.59 MB
Ppersica_139_peptide.fa.gz	Nov 8 2010	6.68 MB
Ppersica_139_readme.txt	Nov 8 2010	2.14 KB
Ppersica_139_transcript.fa.gz	Nov 8 2010	11.18 MB

Figure 3.2: Peach Genome (<http://plantgdb.org/PeGDB/>)

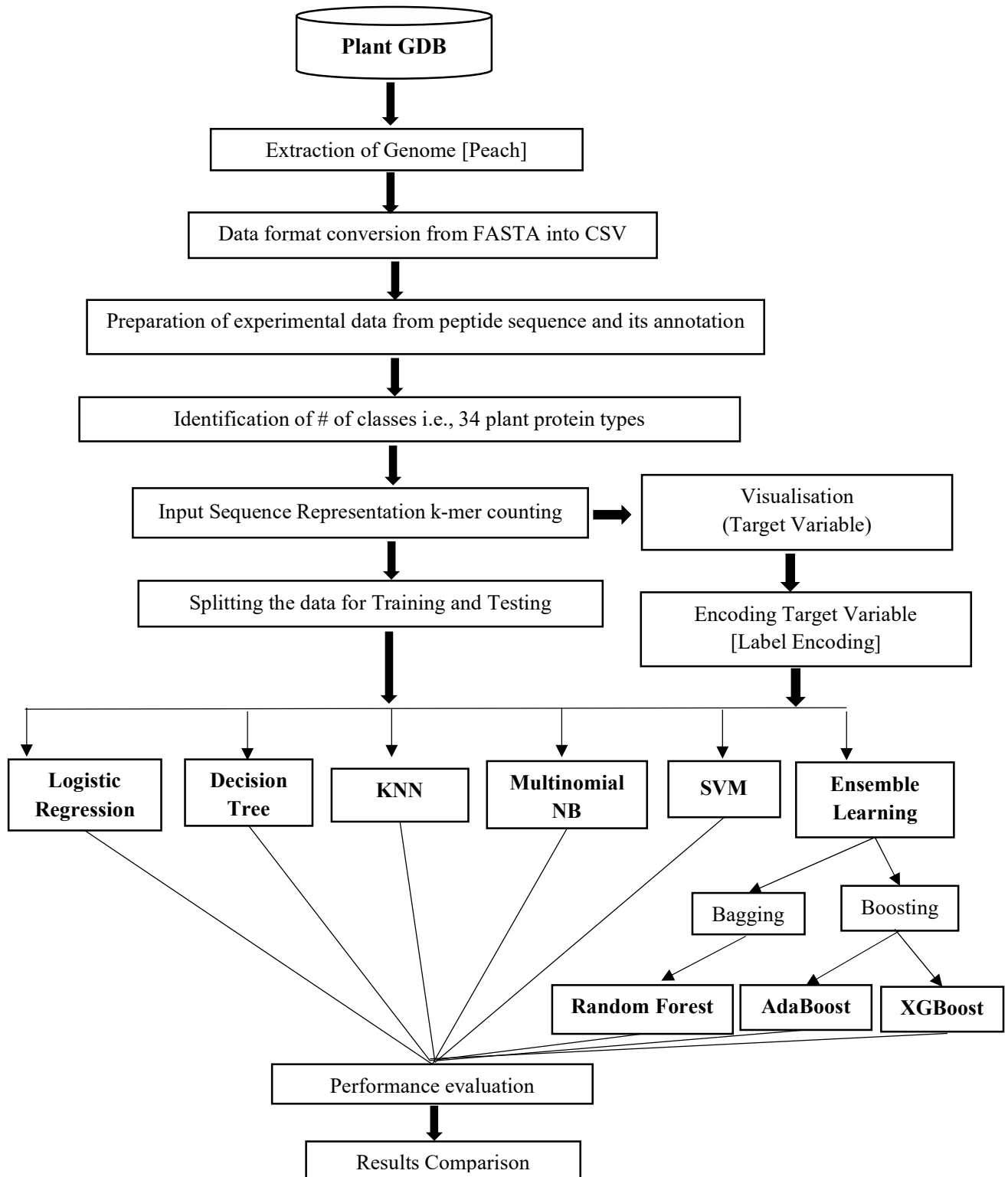


Figure 3.3: Detailed Workflow

3.3 Detailed Architecture

3.3.1 Dataset Preparation

- The Peptide file considered for preparing data is in FASTA format and it needs to be changed to CSV format for applying Machine Learning algorithms. For reading the FASTA file SeqIO.parse() function from BioPython Library is used. Extracted required attributes, stored them in a pandas data frame, and finally saved them in a file named 'peptide_data.csv' as shown in Fig 3.4.

```

1 >ppa010779m|PACId:17640292
2 MAASNSGTVSLKLLIDTTRNKVFAEAGKDFVDFLFTLLSLPAGTIIRLLSKDAMVGSGLKLYESVETLNDEYLQPNLNK
3 DTLLKPKEPVAAGPNLLGLLTDVKSDAPKTIYRCSNGSCIYRISYVADDPKAICPGCKHSMTTAVTYVASPTSEVQATYS
4 GAGKVGYYKGLVTYIMDDLEVKPLSTISCITLLSRFSVKDVGVLEEKVVDLGMDEGVKLLKASLQTKSVLTQVFLR*
5 >ppa004704m|PACId:17640293
6 MSSHLYGYGATQSAAAAAAATAGLSSVYTSRTLTDPTLRYLSGSDPFASATDHHSSSSMYLATSHLMSQSSWPAPDVEP
7 GVPGVKRPSEALYHQSFMGAYNTIGQEAWSALAKRPRYESASNLPIYQRPGEKDCAHYMLTRTCKFGELCKFDHPHIVV
8 PEGGIPDWKEVPLVAPSESLPERPGEPCPYFIKTQRCKFGMRCKFNHPKEKLAASAVASENADVAFALPERPSEPPCAFYM
9 KTGQCKFGATCKFHHPKDIQIPSAEQENKIGETGTTIQPEGTGFAVKLPVSFSPALLYNSKELPVRPGEPCPYFLKTGS
10 CKYGATCRYNHPDRYAINPPIGAISHPIVAPPAAGLNIGVINPAASIYQTLAQPTVGGGQTVYPQRFQGQIECDYYMKTGE
11 CRFGEQCKYHHPIDRSVTLSTTKPVQQNVKLTLAGLPRREGVAICVYYLKTGTCKYGATCKFDHPPPEVGMMAASQG
12 ASGGEANDFTEHQQ*

```



	ID	PACId	Sequence
0	ppa010779m	17640292	MAASNSGTVSLKLLIDTTRNKVFAEAGKDFVDFLFTLLSLPAGTI...
1	ppa004704m	17640293	MSSHLYGYGATQSAAAAAAATAGLSSVYTSRTLTDPTLRYLSGSDP...
2	ppa025459m	17640294	MDQQYSRDFPILSYILSRDPESNPPLSPQLQETLLTQLPHLNHPK...
3	ppa025141m	17640295	MAMKQPHVIFPFPLQGHMKPLLCLAEELLCHAGLHVTVYNTHHNH...
4	ppa001905m	17640296	MCSSTNVPRWTPSPSPTRSLLASAVEGPSSKESHVLDHHAEMKRQ...

Figure 3.4: Peptide File Conversion (.fa to .csv)

- The annotation file was in text format(tab-separated). Read the file, convert it into pandas dataframe, and finally save it to a CSV file named 'annotation_info.csv' as shown in Fig 3.5.

```

ppa000001m    PF07728,PF07726 PTHR22908    KOG1808    AT1G67120.1
              ATP binding / ATPase/ nucleoside-triphosphatase/ nucleotide binding / transcription
factor binding
ppa000002m    PF02207,PF00569 PTHR21725    KOG1776    AT3G02260.1    BIG
              BIG (BIG); binding / ubiquitin-protein ligase/ zinc ion binding
ppa000003m    PF00097 PTHR12183                    AT5G23110.1    zinc
              finger (C3HC4-type RING finger) family protein
ppa000004m    PF00169 PTHR16166    KOG1809    AT4G17140.2
              unknown protein
ppa000005m    PF00169 PTHR16166    KOG1809    AT1G48090.1
              phosphoinositide binding

```




	Transcript Name	PFAM	Panther	KOG	KEGG ec	KEGG Orthology	Name	Symbol	Defline
0	ppa000001m	PF07728,PF07726	PTHR22908	KOG1808	NaN	NaN	AT1G67120.1	NaN	ATP binding / ATPase/ nucleoside-triphosphatas...
1	ppa000002m	PF02207,PF00569	PTHR21725	KOG1776	NaN	NaN	AT3G02260.1	BIG	BIG (BIG); binding / ubiquitin-protein ligase/...
2	ppa000003m	PF00097	PTHR12183	NaN	NaN	NaN	AT5G23110.1	NaN	zinc finger (C3HC4-type RING finger) family pr...
3	ppa000004m	PF00169	PTHR16166	KOG1809	NaN	NaN	AT4G17140.2	NaN	unknown protein
4	ppa000005m	PF00169	PTHR16166	KOG1809	NaN	NaN	AT1G48090.1	NaN	phosphoinositide binding

Figure 3.5: Annotation File Conversion (.txt to .csv)

- The problem statement lies in classifying amino acid sequences into protein families. Extract relevant features such as Sequence and ID columns from 'peptide_file.csv' and Protein families (Defline) and Transcript name columns from 'annotation_file.csv'. Merge these two files to make a single dataset named 'filtered_data.csv' based on common identifiers present in both files (ID and Transcript name) respectively. Protein class labels were extracted based on their occurrence count of at least 50 counts (count $\geq$ 50), the required condition to choose labels, and their corresponding sequence and id. The final dataset named 'filtered_data.csv' consists of 3824 sequence samples over 34 different protein classes as shown in Table 1[3][4].

	ID	Sequence	Defline
0	ppa000003m	MESPVATPESIFLEDGQKVYLTRRIREVLVNYPEGTTVLKELIQN...	zinc finger (C3HC4-type RING finger) family pr...
1	ppa000015m	MATLSQAQAVKSLNKSPGRRRFVFKSFSQRLEEVEIDVFRSLDKVK...	binding
2	ppa000021m	MPDVDLPSAVDPTHHLPLQLPSDPTPPAPTRSDPPGCTLDWLPDF...	transducin family protein / WD-40 repeat famil...
3	ppa000041m	MGTEPALLVDIIFKTLIYDDRGSRKAVDDIITKGLQEVAFMKSF...	binding
4	ppa000048m	MKAGSAAKLIVDALLQRFLPLARRRIETAQAQDGOYLRLPSDPAYEQ...	binding

Figure 3.6: Final Dataset (filtered_data.csv)

Table 3.1: Class Label Description

S.No	Class Label	Significance
1	'C2 domain-containing protein'	Membrane Targeting
2	'DNA binding'	Homologous recombination and Genome maintenance
3	'DNAJ heat shock N-terminal domain-containing protein'	Chaperone binding, heat shock protein binding
4	'F-box family protein'	Responsible for vegetative and reproductive growth
5	'FRS5 (FAR1-related sequence 5); zinc ion binding'	Regulate DNA transcription
6	'GDSL-motif lipase/hydrolase family protein'	Lipid metabolism
7	'RNA recognition motif (RRM)-containing protein'	Binding RNA
8	'UDP-glucuronosyl/UDP-glucosyl transferase family protein'	Involved in UDP glucosyl transferase activity
9	'Ankyrin repeat family protein'	Involved in mediating protein-protein interactions
10	'Aspartyl protease family protein'	Protein Degradation
11	'Basic helix-loop-helix (bHLH) family protein'	Transcription regulation
12	'binding'	Control Gene Regulation
13	'Disease resistance family protein'	Defense Signaling
14	'Disease resistance protein (CC-NBS-LRR class), putative'	Pathogen Recognition
15	'Disease resistance protein (NBS-LRR class), putative'	Immune response
16	'Disease resistance protein (TIR-NBS-LRR class), putative'	Defense activation

S.No	Class Label	Significance
17	'hAT dimerization domain-containing protein'	Gene expression Regulation
18	'Hydroxyproline-rich glycoprotein family protein'	Cell wall modification
19	'Leucine-rich repeat family protein / protein kinase family protein'	Regulates shoot and root growth
20	'Leucine-rich repeat transmembrane protein kinase, putative'	Innate immunity in plants
21	'Nucleic acid binding'	Gene regulation
22	'Nucleic acid binding / ribonuclease H'	Catalyse RNA cleavage
23	'Oxidoreductase, 2OG-Fe (II) oxygenase family protein'	Protein and Nucleic acid modification
24	'Pentatricopeptide (PPR) repeat-containing protein'	Editing RNA
25	'Protein binding / zinc ion binding'	Regulate growth and stress adaptation in plants
26	'Protein kinase family protein'	Respond to environmental stresses
27	'Protein kinase, putative'	Regulate biological activity
28	'Reverse transcriptase, putative / RNA-dependent DNA polymerase, putative'	Responsible for transcriptional and post-transcriptional gene silencing
29	'Short-chain dehydrogenase/reductase (SDR) family protein'	Metabolic regulation
30	'Transducin family protein / WD-40 repeat family protein'	Nucleotide-binding
31	'Transferase family protein'	Involved in transferase activity
32	'Zinc finger (C3HC4-type RING finger) family protein'	ATP and Zinc ion binding
33	'Zinc finger protein-related'	Regulate growth and Stress adaptation
34	'Zinc knuckle (CCHC-type) family protein'	Nucleic acid and zinc ion binding

3.4. K-mer Encoding

When processing the Genome sequence, conversion from string format to numerical value is necessary, to form a matrix input for model training. The features of the existing sequence encoding methods are shown in Table 2.

Table 3.2: Sequence Encoding Methods

Encoding Method	Features
Sequential encoding	Encodes each character into a numeric value.
One-hot encoding	Represents the categorical to binary value mapping in a binary vector format.
K-mer encoding	Divide the sequence into k-length overlapping short sequences or segments

The problem that exists in the methods(other than k-mer encoding) mentioned above in Table 2 does not result in vectors of uniform length, which is the required condition to feed data to an algorithm (classification or regression). For the other two methods, we have to curtail or fill with “n” or “0” to meet the requirement.

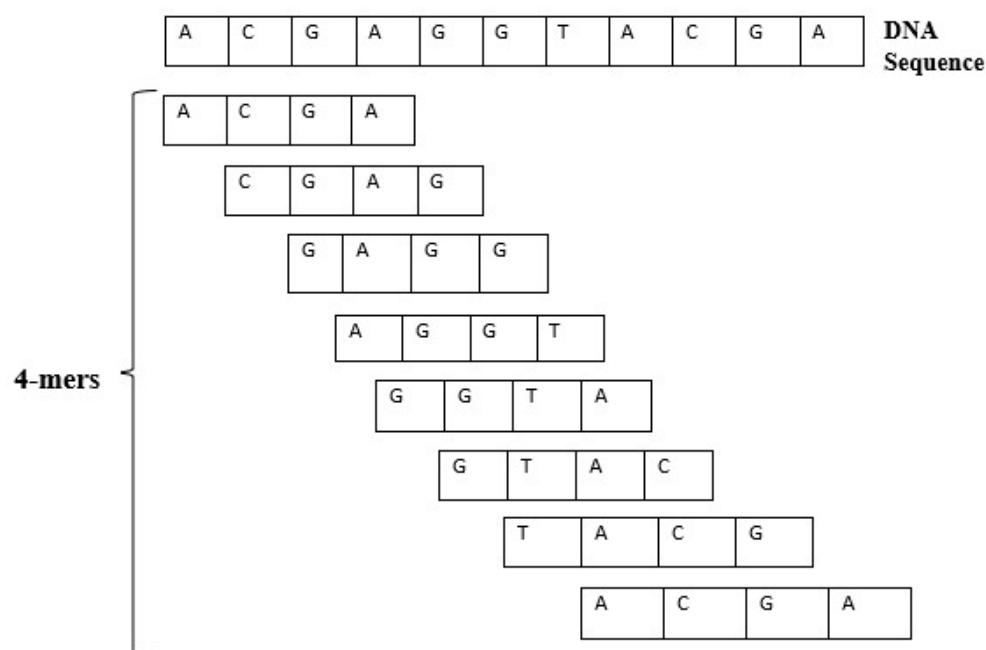


Figure 3.7: 4-mer representation

3.5 Algorithms Used

Algorithm 1: Label Encoder

Input: Target Variable (y)

Output: Encoded Target Variable (y_encoded)

Method

Import the necessary libraries

Initialize an instance of the LabelEncoder class as le

for each y

 Call the fit_transform () method on the le object, passing y as the argument, and assign the transformed values to a new variable

 y_encoded=le.fit_transform(y).

return the encoded target variable (y_encoded).

Algorithm 2: K-mer Technique

Input: Amino acid string sequence

Output: Numeric N-gram representation (N=4)

Method

Divide the amino acid sequence into overlapping segments of length K (k=6).

//create the BOW model using CountVectorizer()

Convert list of kmers into string sentences of words

for each n-gram

 count the frequency

return the document-term matrix representation.

Algorithm 3: Logistic Regression

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_1

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the logistic regression model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C_1 model using the testing data.
- Assess the model's performance.

Algorithm 4: Decision Tree

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_2

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.

- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the Decision Tree model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C2 model using the testing data.
- Assess the model's performance.

Algorithm 5: K-Nearest Neighbor

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_3

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the KNN model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C3 model using the testing data.
- Assess the model's performance.

Algorithm 6: Multinomial Naive Bayes

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_4

Method

- Encode Sequence Data using the K-mer Technique.

- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the Multinomial NB model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C4 model using the testing data.
- Assess the model's performance.

Algorithm 7: Support Vector Machine

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_5

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the SVM model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C5 model using the testing data.
- Assess the model's performance.

Algorithm 8: Random Forest

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_6

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the Random Forest model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained C_6 model using the testing data.
- Assess the model's performance.

Algorithm 9: AdaBoost

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_7

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.

- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the AdaBoost model with the desired hyperparameters.
- Training the classifier..
- Evaluate the trained $C7$ model using the testing data.
- Assess the model's performance.

Algorithm 10: XGBoost

Input:

- A set of protein sequence, $P = \{p_1, p_2, \dots, p_m\}$ where sequence is represented as p_i , $1 \leq i \leq m$ and $m = |P|$ with its corresponding *label* represented as L_j , $1 \leq j \leq 34$.

Output:

- Classifier Model C_8

Method

- Encode Sequence Data using the K-mer Technique.
- Encode Protein labels using a label encoding technique.
- Data Splitting for Training and Testing Purposes ($D_{\text{Train}} = 80$, $D_{\text{Test}} = 20$).
- Initialize the XGBoost model with the desired hyperparameters.
- Training the classifier.
- Evaluate the trained $C8$ model using the testing data.
- Assess the model's performance.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Evaluation Metrics

To examine the ML model's performance, evaluation metrics are used. Below are the few evaluation metrics considered for this classification task.

4.1.1 Confusion Matrix

It is a table that represents the following four values: TPs, TNs, FPs, and FNs. It is used in examining the model's performance and identify the specific errors or misclassifications in protein labeling.

	0	1
0	TP	TN
1	FP	FN

4.1.2 Accuracy

Measures the overall correctness of the model's predictions for protein labels.

$$\text{Accuracy} = \text{TP} + \text{TN} / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad \text{eq 4.1}$$

4.1.3 Precision

Measures the proportion of correctly predicted positive protein labels out of all samples that are predicted as positive, the focus is on positive prediction accuracy.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad \text{eq 4.2}$$

4.1.4 Recall

Measures the proportion of correctly predicted positive protein labels out of actual instances of a specific protein.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad \text{eq 4.3}$$

4.1.5 F1-score

Measures the model's accuracy considering both precision and recall (harmonic mean).

$$\text{F1-score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall}) \quad \text{eq 4.4}$$

4.2 Experimental Evaluation Using Entire Dataset (with 34 classes)

The dataset used to train is “filtered_data.csv” prepared by taking files from [1] database. Detailed dataset preparation is mentioned in section 3.3.1. The dataset consists of 3824 sequence samples over 34 different protein classes as shown in Table 1. For implementation python language is used and 8 Machine learning algorithms were considered which were explained in section 3.5.

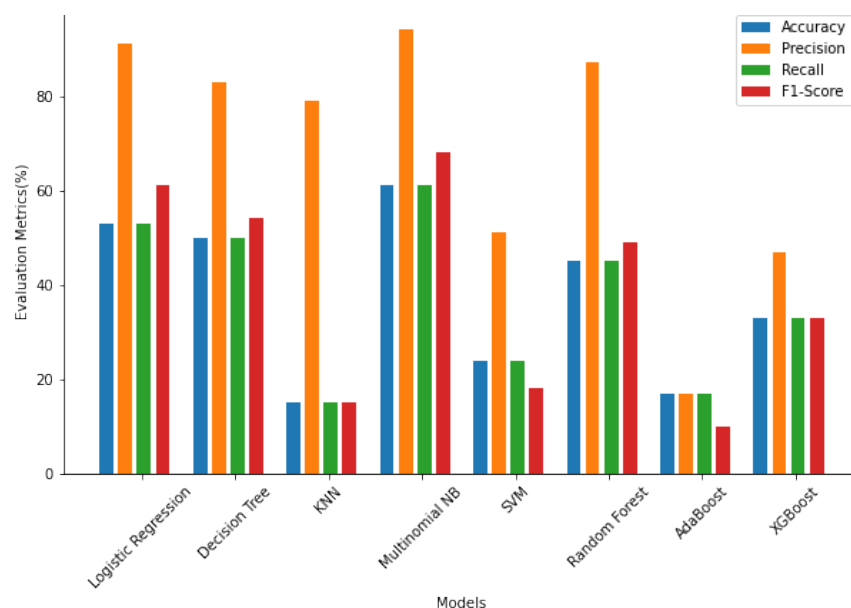
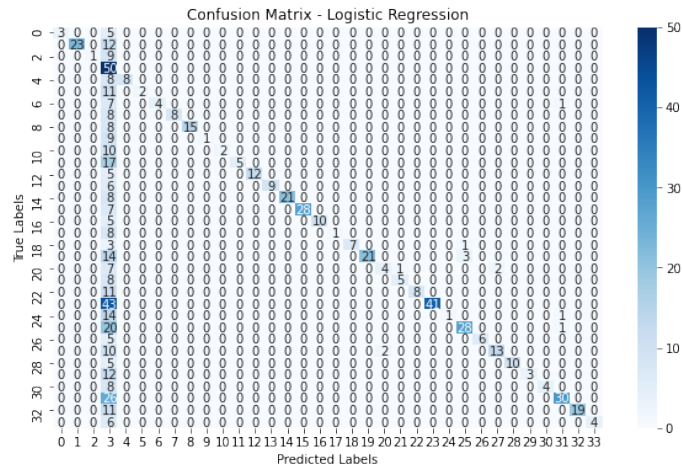
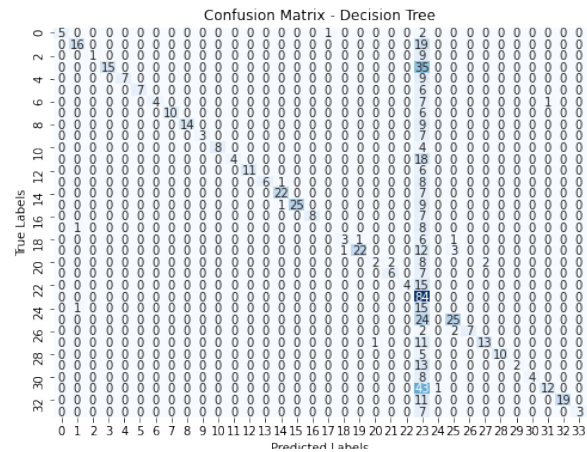


Fig 4.1 Evaluation metric scores of all models

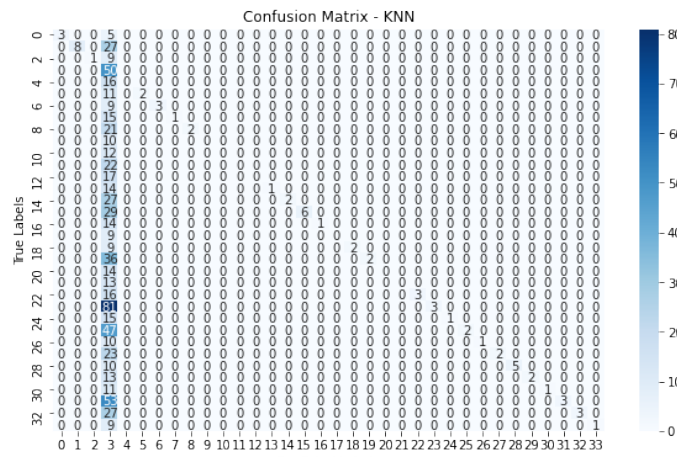
From various plots obtained as shown from above figures, it is shown that Multinomial Naïve Bayes outperformed other methods.



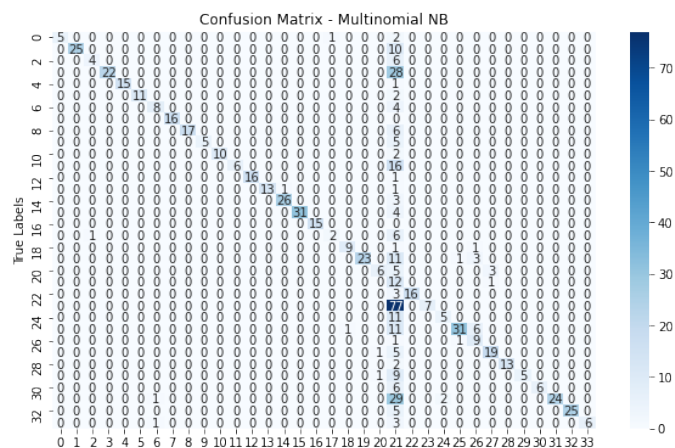
(a) Confusion Matrix - LR



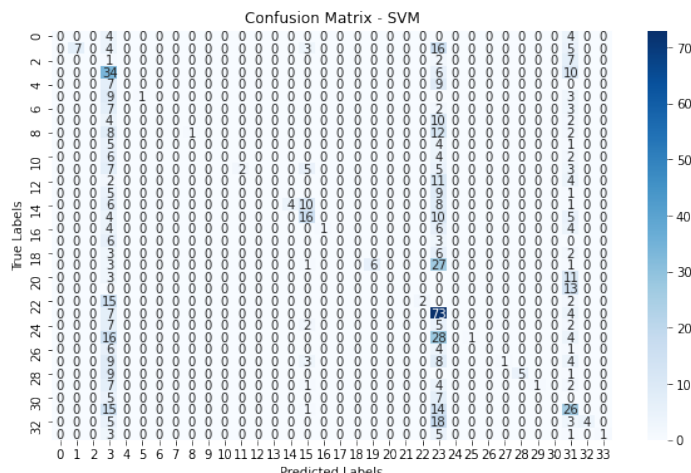
(b) Confusion Matrix - DT



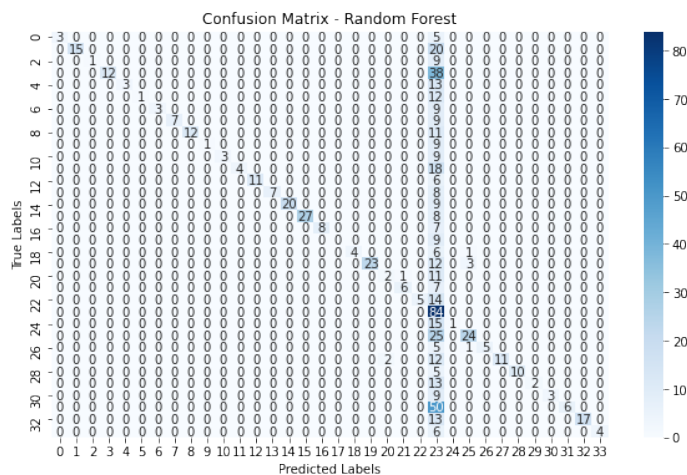
(c) Confusion Matrix - KNN



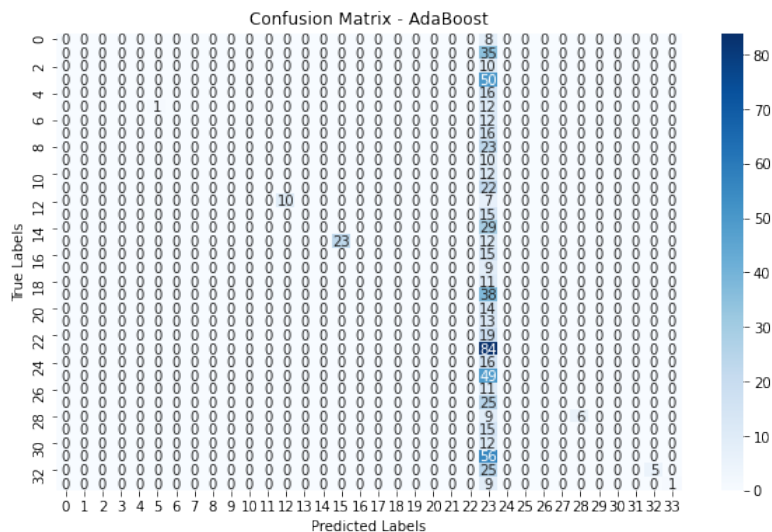
(d) Confusion Matrix – Multinomial NB



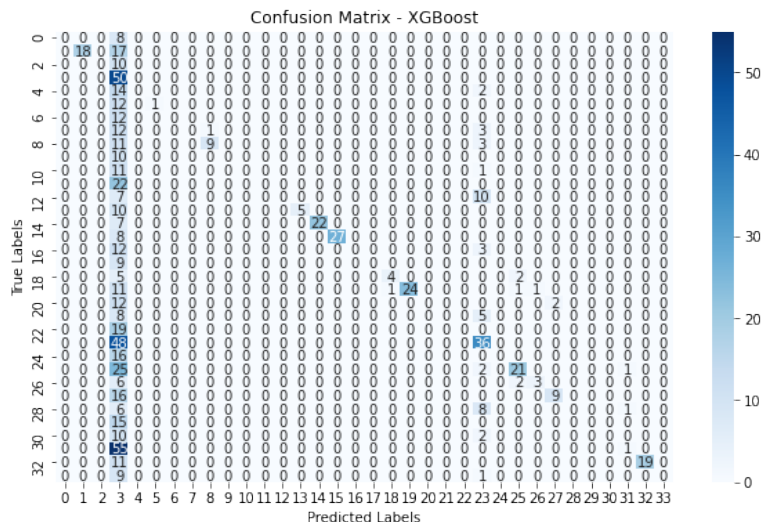
(e) Confusion Matrix - SVM



(f) Confusion Matrix - RF



(g) Confusion Matrix - AdaBoost



(h) Confusion Matrix - XGBoost

Fig 4.2 Confusion Matrix of all classes

The amount of true positives, true negatives, false positives, and false negatives are displayed in the confusion matrices for each of these models, providing a detailed analysis of how effectively they were able to categorise plant proteins using k-mer encoding. With the use of these matrices, different models' performance in this classification job may be assessed and compared.

4.3 Divided Datasets Evaluation

From confusion matrix plots of various classifiers, it is clearly shown that classes 21, 23, and 3 were assigned to most of the instances. Therefore, the dataset is split into two parts and was trained separately such that one dataset comprises classes 21, 23, and 3. And other dataset comprises of remaining 31 classes. The results from the two different datasets were shown below.

Dataset 1(D1) - [3: 'F-box family protein', 21: 'nucleic acid binding/ribonuclease H', 23: 'pentatricopeptide (PPR) repeat-containing protein']

Dataset 2(D2) – ['Remaining 31 labels']

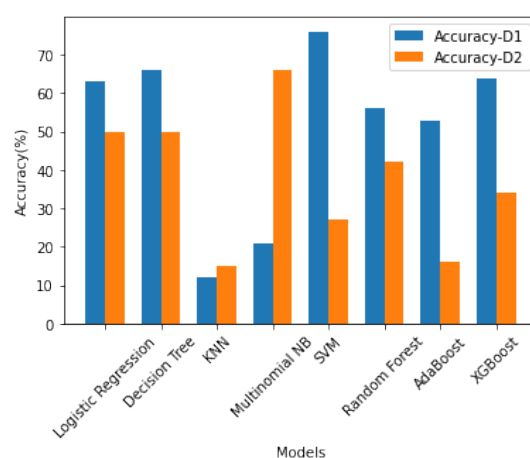


Fig 4.3 Accuracy plots [D1] [D2]

Accuracy plots show how each model's classification accuracy changes as a function of different parameters or hyperparameter settings. They aid in identifying the ideal setups for improving the precision of plant protein classification using k-mer encoding, making it feasible to select the most appropriate model for the task with knowledge.

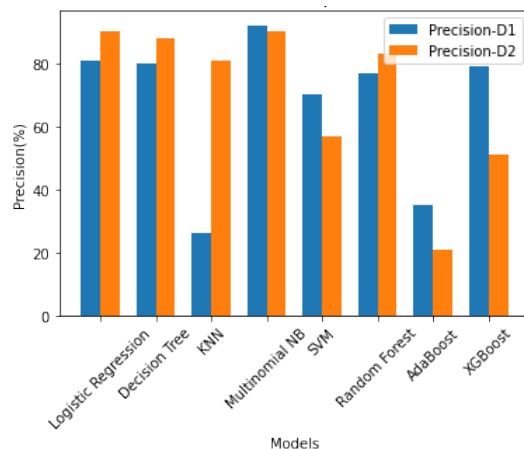


Fig 4.4 Precision plots [D1] [D2]

The precision scores for each model are displayed on precision plots as a function of the parameters or hyperparameter settings. They aid in identifying the best setups for increasing the precision of plant protein classification by k-mer encoding, enabling informed decisions about the most suitable model for the task at hand.

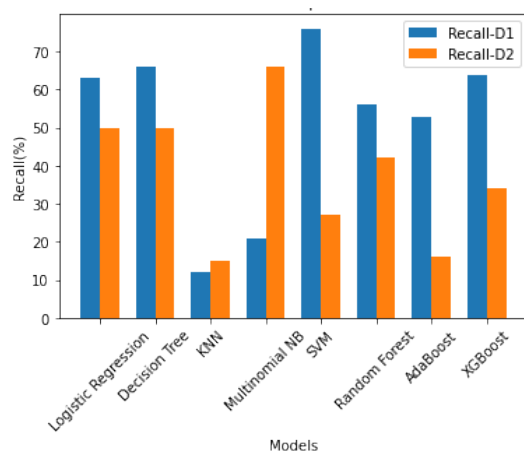


Fig 4.5 Recall plots [D1] [D2]

Recall plots display how various parameter or hyperparameter settings affect each model's recall scores. It is feasible to choose the most appropriate model for the task by employing them to help determine the optimum configurations for getting higher recall in plant protein categorization using k-mer encoding.

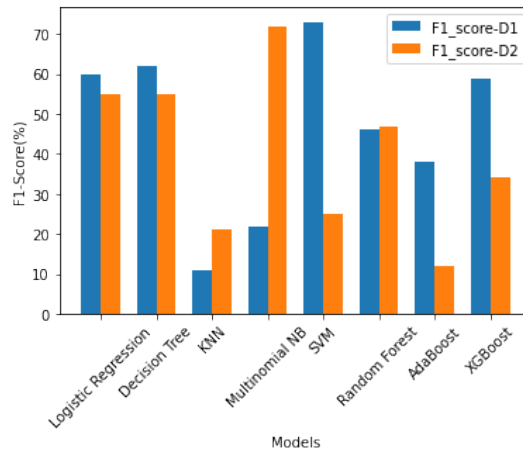


Fig 4.6 F1-score plots [D1] [D2]

F1-score graphs demonstrate the effects of changing parameters or hyperparameter settings on the F1-scores of each model. They aid in figuring out the best setups for getting higher F1-scores when classifying plant proteins using k-mer encoding, making the choice of a model more informed.

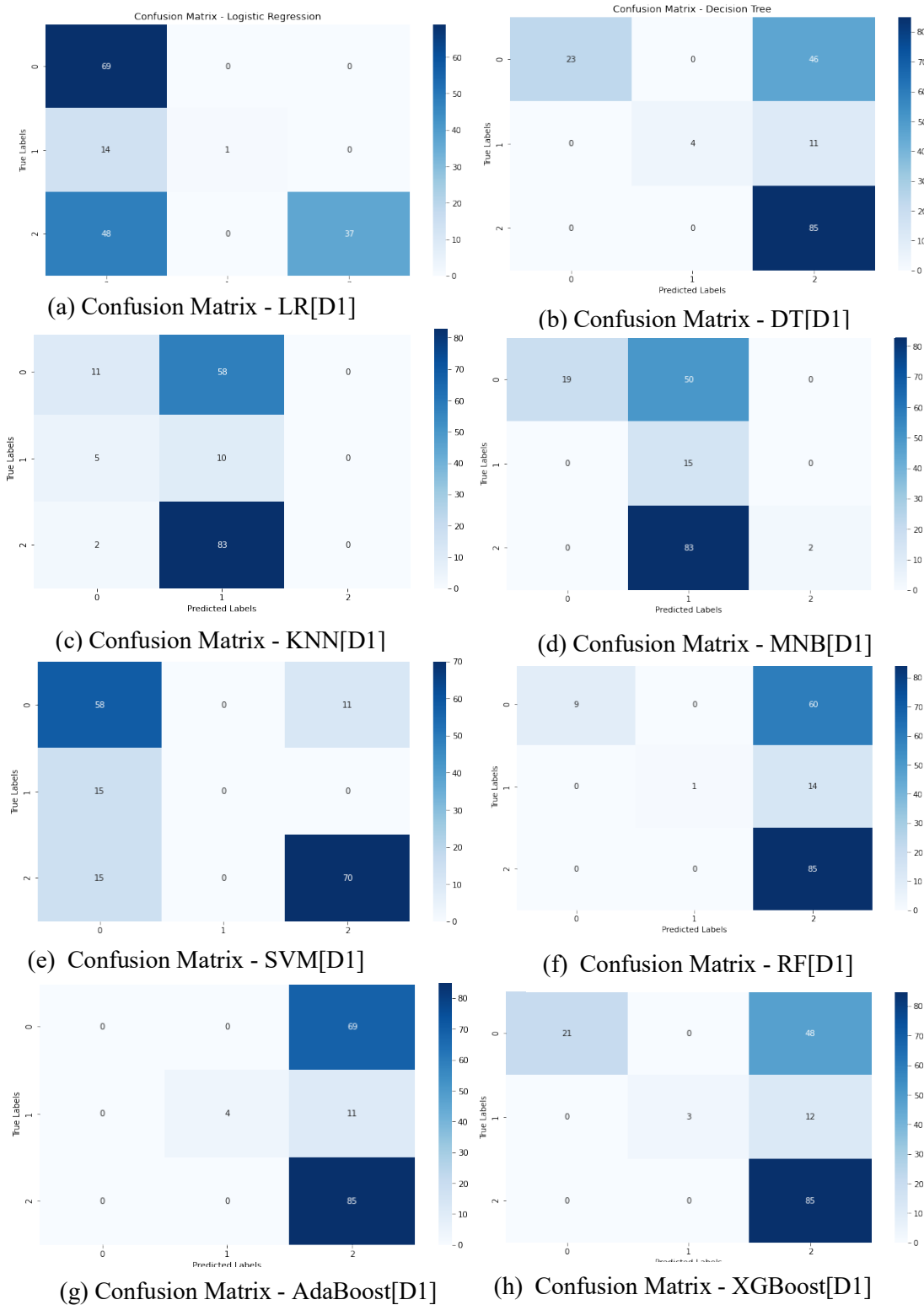
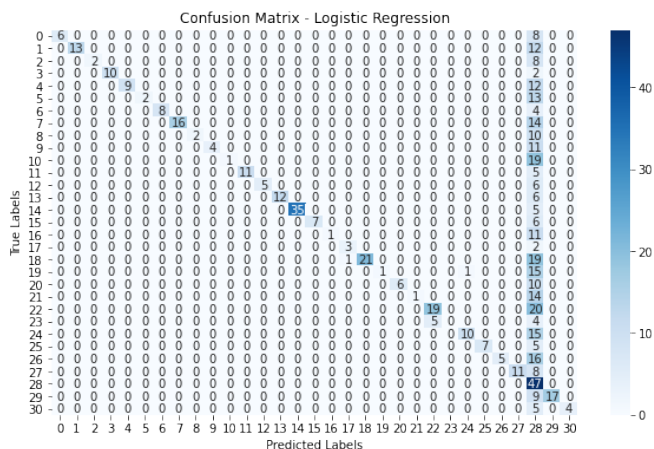
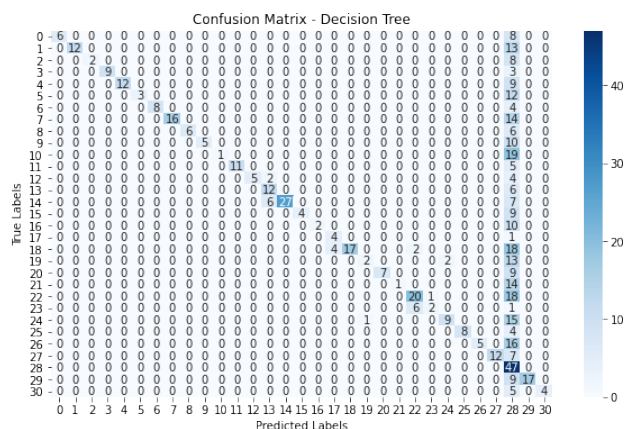


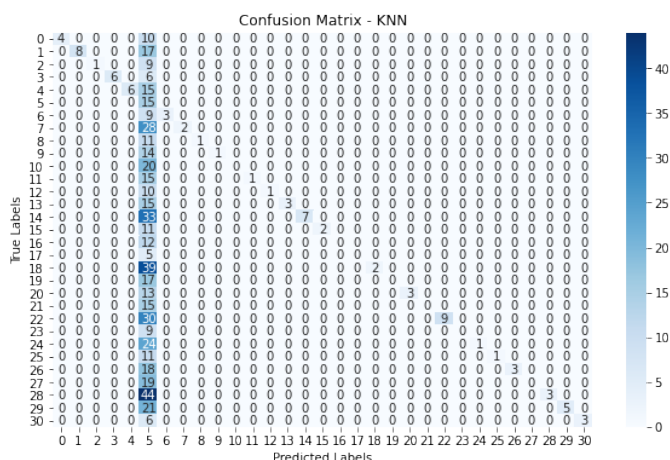
Fig 4.7 Confusion Matrix of all classes[D1]



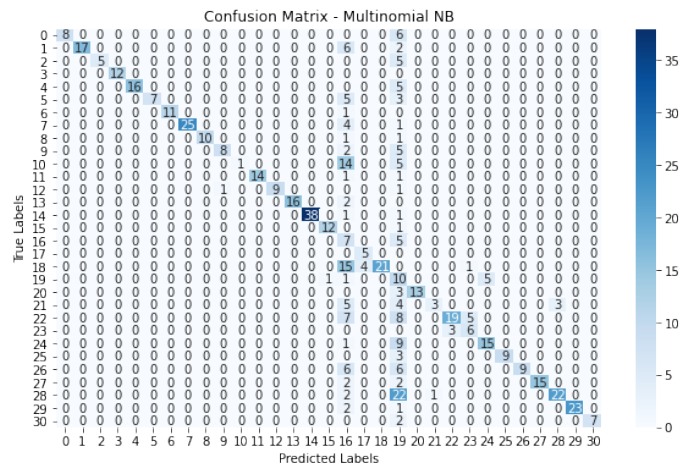
(a) Confusion Matrix - LR[D2]



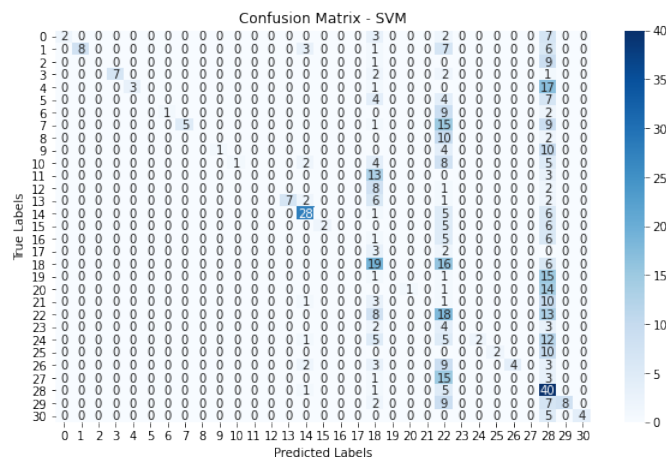
(b) Confusion Matrix - DT[D2]



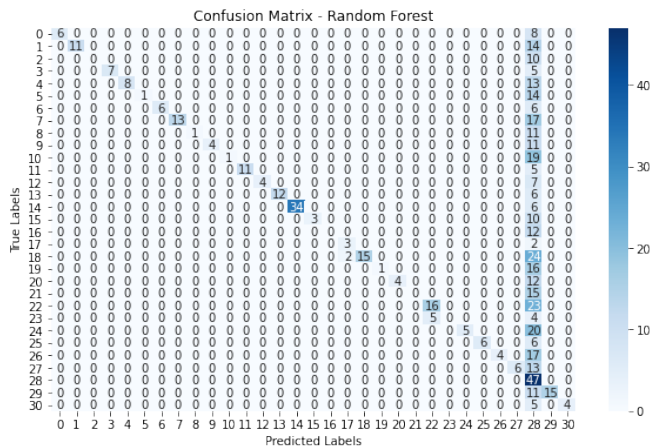
(c) Confusion Matrix - KNN[D2]



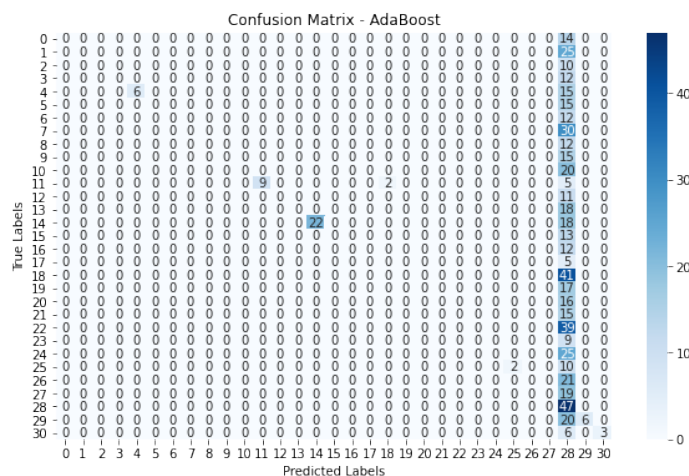
(d) Confusion Matrix - MNB[D2]



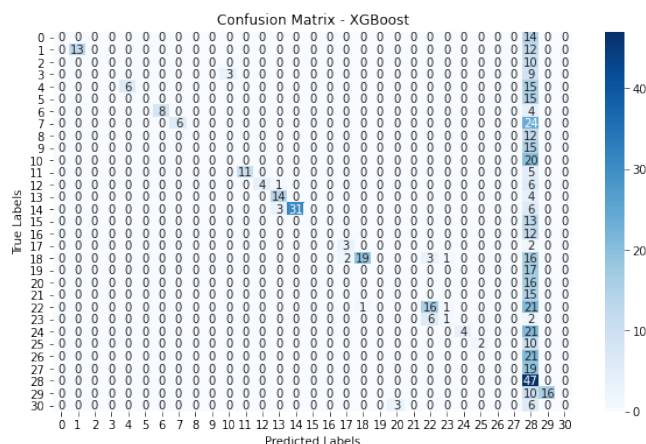
(e) Confusion Matrix - SVM[D2]



(f) Confusion Matrix - RF[D2]



(g) Confusion Matrix - AdaBoost[D2]



(h) Confusion Matrix - XGBoost[D2]

Fig 4.8 Confusion Matrix of all classes[D2]

The confusion matrices on Datasets 1 and 2 for each model provide a thorough breakdown of each model's performance in the k-mer encoding-based categorization of plant proteins. They help with comparing and evaluating the efficacy of various models on diverse datasets by showing the amount of true positives, true negatives, false positives, and false negatives for each example.

Evaluation Metric Results:

Table 4.1 Evaluation Metric Results

Model Metric (%)	Accuracy	Precision	Recall	F1-score
Logistic Regression	53	91	53	61
Decision Tree	50	83	50	54
KNN	15	79	15	15
Multinomial NB	61	94	61	68
SVM	24	51	24	18
Random Forest	45	87	45	49
AdaBoost	17	17	17	10
XGBoost	33	47	33	33

Table 4.2 Evaluation Metric Results – D1

Model Metric (%)	Accuracy	Precision	Recall	F1-score
Logistic Regression	63	81	63	60
Decision Tree	66	80	66	62
KNN	12	26	12	11
Multinomial NB	21	92	21	22
SVM	76	70	76	73
Random Forest	56	77	56	46
AdaBoost	53	35	53	38
XGBoost	64	79	64	59

Table 4.3 Evaluation Metric Results – D2

Model Metric (%)	Accuracy	Precision	Recall	F1-score
Logistic Regression	50	90	50	55
Decision Tree	50	88	50	55
KNN	15	81	15	21
Multinomial NB	66	90	66	72
SVM	27	57	27	25
Random Forest	42	83	42	47
AdaBoost	16	21	16	12
XGBoost	34	51	34	34

From Table 4.2 and 4.3 results, it is shown that Support Vector Machine (SVM) outperformed in Dataset 1 and Multinomial NB outperformed in Dataset 2.

CHAPTER 5

CONCLUSION and FUTURE WORKS

This project involves evaluating various classifiers, where Multinomial Naive Bayes (NB) achieved the highest accuracy. Notably, three classes (3, 21, and 23) were consistently assigned to most instances based on the confusion matrices. To delve deeper into this phenomenon, the dataset was divided into two parts: one comprising the frequently assigned classes and the other comprising the remaining classes. In dataset one, Support Vector Machine (SVM) exhibited the best performance, while Multinomial NB outperformed other models in dataset two.

Future work involves optimizing accuracy by exploring additional classifiers such as deep learning models like Convolutional Neural Networks (CNN) and fine-tuning existing models. Additionally, investigating alternative feature representation techniques and incorporating additional biological or contextual features can contribute to enhanced classification performance. The project's outcomes have practical implications in phenotyping, facilitating advancements in crop improvement, disease resistance, and overall plant breeding strategies.

REFERENCES

1. Oncul, A. B., & Celik, Y. (2023). Signal, Image and Video Processing, 17(5), 2055-2061.
2. Yao, Jiepeng, Yi Ling, Peichen Hou, Zhongyi Wang, Lan Huang. A graph neural network model for deciphering the biological mechanisms of plant electrical signal classification. Applied Soft Computing 137 (2023): 110153.
3. Nedyalkova, Miroslava, Sergio Madurga, Marek Tobiszewski, Vasil Simeonov. Sequence-Based Prediction of Plant Allergenic Proteins: Machine Learning Classification Approach. ACS Omega (2023).
4. Upadhyaya, Shriprabha R., Philipp E. Bayer, Cassandra G. Tay Fernandez, Jakob Petereit, Jacqueline Batley, Mohammed Bennamoun, Farid Boussaid, David Edwards. Evaluating Plant Gene Models Using Machine Learning. Plants 11.12 (2022): 1619
5. Juneja, Sapna, Annu Dhankhar, Abhinav Juneja, Shivani Bali. An Approach to DNA Sequence Classification Through Machine Learning: DNA Sequencing, K Mer Counting, Thresholding, Sequence Analysis. International Journal of Reliable and Quality E-Healthcare (IJRQEH) 11.2 (2022): 1-15.
6. Vigil, MS Antony, M. Mirutuhula, Sure Sarvagna, R. Supraja, and Gadusunari Priyanka Reddy. DNA Sequencing Using Machine Learning Algorithms. In 2022 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI), vol. 1, pp. 1-4. IEEE, 2022.
7. Orozco-Arias, Simon, Mariana S. Candamil-Cortés, Paula A. Jaimes, Johan S. Piña, Reinel Tabares-Soto, Romain Guyot, and Gustavo Isaza. K-mer-based machine learning method to classify LTR-retrotransposons in plant genomes. PeerJ 9 (2021): e11456. Wattanapornprom, Warin, et al. "Ensemble of multiple classifiers for multilabel classification of plant protein subcellular localization." Life 11.4 (2021): 293.

8. Gunasekaran, Hemalatha, K. Ramalakshmi, A. Rex Macedo Arokiaraj, S. Deepa Kanmani, Chandran Venkatesan, and C. Suresh Gnana Dhas. Analysis of DNA sequence classification using CNN and hybrid models. *Computational and Mathematical Methods in Medicine* 2021 (2021).
9. Guo, Yuxin, Liping Hou, Wen Zhu, and Peng Wang. Prediction of Hormone-Binding Proteins Based on K-mer Feature Representation and Naive Bayes. *Frontiers in Genetics* 12 (2021): 797641.
10. Zhang, Jinny X., Boyan Yordanov, Alexander Gaunt, Michael X. Wang, Peng Dai, Yuan-Jyue Chen, Kerou Zhang et al. A deep learning model for predicting next-generation sequencing depth from DNA sequence. *Nature communications* 12, no. 1 (2021): 4387.
11. Sangphukieo, Apiwat, Teeraphan Laomettachit, and Marasri Ruengjitchatchawalya. Photosynthetic protein classification using genome neighborhood-based machine learning feature. *Scientific Reports* 10, no. 1 (2020): 7108.
12. Yadav, A. K., & Singla, D. (2020). VacPred: Sequence-based prediction of plant vacuole proteins using machine-learning techniques. *Journal of biosciences*, 45, 1-9.
13. Sekhar, S. M., Siddesh, G. M., Raj, M., & Manvi, S. S. (2021). Protein class prediction based on Count Vectorizer and long short term memory. *International Journal of Information Technology*, 13(1), 341-348.
14. Gupta, C. P., Bihari, A., & Tripathi, S. (2019). Human Protein Sequence Classification using Machine Learning and Statistical Classification Techniques. *Int. J. Recent Technol. Eng*, 8(2).
15. Solis-Reyes, S., Avino, M., Poon, A., & Kari, L. (2018). An open-source k-mer based machine learning tool for fast and accurate subtyping of HIV-1 genomes. *PloS one*, 13(11), e0206409.
16. Phan, D., Ngoc, G. N., Lumbanraja, F. R., Faisal, M. R., Abipihi, B., Purnama, B., ... & Satou, K. (2017). Combined use of k-mer numerical features and position-specific categorical features in fixed-length DNA sequence classification. *Journal of Biomedical Science and Engineering*, 10(8), 390-401.

17. Gotoh, O., Morita, M., & Nelson, D. R. (2014). Assessment and refinement of eukaryotic gene structure prediction with gene-structure-aware multiple protein sequence alignment. *Bmc Bioinformatics*, 15(1), 1-13.
18. <http://plantgdb.org/> - "PlantGDB [Genome Database]" [14/06/2023]
19. <http://plantgdb.org/PeGDB/> - "Prunus persica [Peach] Genome" [14/06/2023]
20. <https://www.uniprot.org/> - "UniProt Database" [14/06/2023]
21. <https://pubmed.ncbi.nlm.nih.gov/> - "PubMed Repository" [14/06/2023]

APPENDIX A

Input Sequence:

MLITAHFIDGEWNLHKRVLNFCVIANHKGNSIGKLLETCLLHWDLKKILAITT
DNTSSNTKAIDYLKSKMGHWKNGSLVLEGKYTHVRCCAHIIVNLIVQDGLKK
LEKSILSIRNAVRYVFYNMTLKMSISLYLASHTTFYNLIAMEGEIGDLFFGDQL
PSQTHSTILKDIACSMKMKFEKYWIDLKVNQLLMVAHVLDPRYLTDPIEK
VIPNFDIFKWWKLNGVKYPGLSLIAKDVLDPSSSTVISESFFAQVVNSFCTFLT
PKIVEALICSQN

Length: 276

Predicted Label: ['DNA binding']

Input Sequence:

MSESSATTA AAAESHAPLLLPRQASSGRDSSDLGSPTARPATLAMLLGRAT
GRRGPSMLVRETAARELDERRADWGYSKPVVALDMMWNTAFVVVSVAML
IWTDEHPNTPIRLWICGYALQCFVHVVLVWVEYQRRNNIARRISRRNQDAQ
QNQHQVEIDANDTDDEEAAAAAARDLAISTRSSLKRCETVNTMASFLWWIV
GFYWVWAGGEVLLQNAPRLYWLAVVFLAFDVFFAIFCVVLACLIGVALCCC
LPCIIAILYAVAGQEGASDADLSILPKYRFRSSSEEKSSVGAGKMIPVETSSGY
LAIERILLPEDAECCICLSPYEDGTELHTLPCNHHFHATCIVKWLKMNATCPL
CKYNILKGNEPV

Length: 374

Predicted Label: ['zinc finger (C3HC4-type RING finger) family protein']

Multinomial Naive Bayes Predictions

Input Sequence:

MAFPEHLTADILSRLPAKSLVRFSSVRKEWQALIRDPHFISQHLNLNIAKEDQS
 GRLLVKYRCNVTRKNVISLLLHGTHGYMLSKLEIPSSVMDSHHLKLVGSCNG
 LLCLAHDKSAHLWSPMILLWNPAIREFSVLPDSVIDSVNDDDNVQARGIAHG
 FGYHPLIDDYKVVRIVSLSCFGNAWIRAEVYSSKMDCWQEVNTFHFEIYEHS
 CTALNGLLYWIAYGKGDRELIVSFDMCDDVFGELQLPDLSFLDTPMCTKLAE
 LKQSLYLIVYTFSGRQREIYVWKMIEHDAEVFWKREWTIGPLLGIERPLGCGL
 YGEIYMESRKGDVLVYNPSSEEAKHIPICGLRCSLEVHFYVESLVSTKGQREV
 LQETNKRIM

Length: 377

Predicted Label: ['F-box family protein']

Input Sequence:

MVSRQIQLLIQRSSSAHLLQLHSLVLKTGLDHNPNFVSHYIYAACSISVEFSK
 LVFDHIPIKPPLFAWNSIIRAFTNSSNPLESNFFFQLHRIGLKPDNFTYPFVLK
 ACGRCSIVGVGGPVHSLILKVGFDSDRYIRNTLLRMYAACGAIRLARRVFDE
 MTVRDVVSWSMIAGYVACNCPLDAFKVFQDMKLANENPNVTLVSLLSAC
 TRLLNFRAGESIHCIYVNCIGIDVALGTALLEMYSKCGHVEKAFQVFNSMSE
 KNLQSWTIMISGLPDHGHGKDAISLFTRMEQTGLVPDSMSFSAILSACSHLGL
 VHKGQRQFFNQMVRTYGIQPTLEHYGCLVDLLGRAGLIEEAYEIIKNMPMEPN
 SVILRSFLCACRNHGVAITLDDKLRKLLISEPNLGANYVLAASVSSLSGCWND
 AADLRVAMKQNGLVKVPGCWSVEVGGGGSTAATINEPLV

Length: 462

Predicted Label: ['pentatricopeptide (PPR) repeat-containing protein']

SVM Predictions - Dataset 1(3 Classes)

Input Sequence:

MFLRQAAANLIKRRFRLPSQTVSSTSGTTTTIPLSSRTKFSYQYHGPKDFSWRM
NYSHRDPRVWLFSLGNAAIIFTIHSNTVLAKDASIESSPENDTEGAEAMGLRKI
EDGSVISNIHTSKWRVFTDKGRDLFLQERILCLHSFGKLEEA EKLFLLALQEA
KEGFGERDPHVASACNNLAELYRVNKAYNKAEPYLYDAISILEESFGPEDIRIS
THSLFLYYL

Length: 223

Predicted Label: ['nucleic acid binding']

Input Sequence:

MVQILHISTTTVRPTTQDDQLTAPKIELTPWDLRLIQFDYIQKGILFHKSAETE
DSNVSLVQRLKDSLTLNIFYPLAGRLVVSENAEDSTVSISIKCNGAGAEFV
HAAADGVKVADILNPYIPDHVIFNLFSLNEVRNYEGVSKPLLSAQVTELLDGI
FIGCSLNHLVADGTSFWHFFNTWAEISRSGSHKILQPRPVFGRQFLDGVIDLPV
HLPFSYSEIIGEPIIKSSSFNAFLQRVVFHFPKEKIAKLKAKANAEMGTTSNISSL
QALMAHLWQATTRARHLQPDQQVNYRFAIGLRQRWEPETLPKEYLGTAIIG
VNVTAIASSELLQNGLGWAASQINKKISSLTPEQARNDLVDWVNKPTLIPNMR
NPTLLARLLTGSSPRFNVYGNDFGWGRPLAVRSGAGDKMDGKLTVPGAEE
GSIDFEACLLPETIQTMMDDAEFMEAVA

Length: 452

Predicted Label: ['transferase family protein']

Multinomial NB Predictions - Dataset 2